

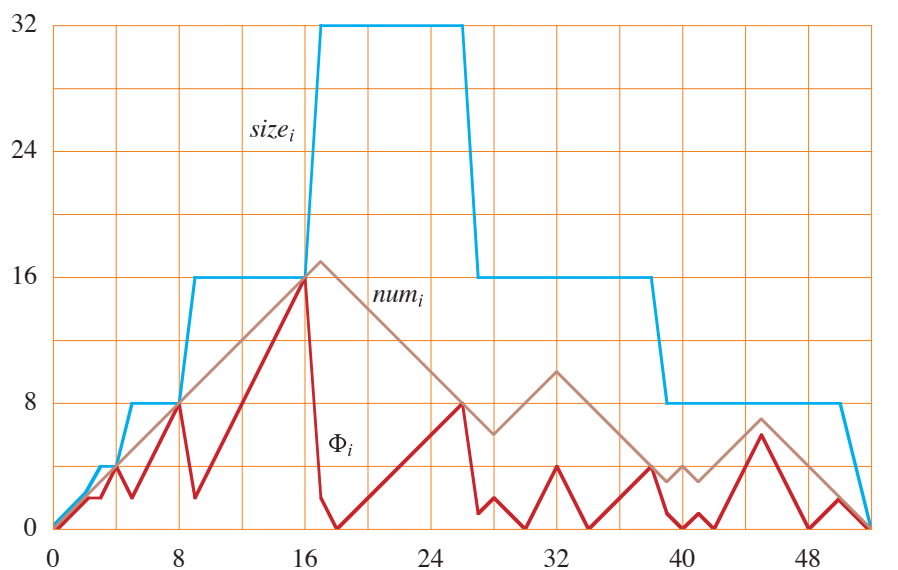


Figure 16.6 The effect of a sequence of n TABLE-INSERT and TABLE-DELETE operations on the number num_i of items in the table (the brown line), the number $size_i$ of slots in the table (the blue line), and the potential (the red line)

$$\Phi_i = \begin{cases} 2(num_i - size_i/2) & \text{if } \alpha_i \geq 1/2, \\ size_i/2 - num_i & \text{if } \alpha_i < 1/2, \end{cases}$$

where $\alpha_i = num_i / size_i$, each measured after the i th operation. Immediately before an expansion or contraction, the potential has built up to the number of items in the table, and therefore it can pay for moving all the items to the new table.

- if the i th operation is an insertion, its amortized cost $\hat{c}_i$ is $c_i + \Delta\Phi_i$, which is $1 + 2 = 3$ if the load factor stays at or above $1/2$, and $1 + (-1) = 0$ if the load factor stays below $1/2$, and
- if the i th operation is a deletion, its amortized cost $\hat{c}_i$ is $c_i + \Delta\Phi_i$, which is $1 + (-2) = -1$ if the load factor stays at or above $1/2$, and $1 + 1 = 2$ if the load factor stays below $1/2$.

Four cases remain: an insertion that takes the load factor from below $1/2$ to $1/2$, a deletion that takes the load factor from $1/2$ to below $1/2$, a deletion that causes the table to contract, and an insertion that causes the table to expand. We analyzed that last case at the end of Section 16.4.1 to show that its amortized cost is 3.

When the i th operation is a deletion that causes the table to contract, we have $num_{i-1} = size_{i-1}/4$ before the contraction, then the item is deleted, and finally $num_i = size_i/2 - 1$ after the contraction. Thus, by equation (16.5) we have

$$\begin{aligned}
\Phi_{i-1} &= \text{size}_{i-1}/2 - \text{num}_{i-1} \\
&= \text{size}_{i-1}/2 - \text{size}_{i-1}/4 \\
&= \text{size}_{i-1}/4,
\end{aligned}$$

which also equals the actual cost c_i of deleting one item and copying $\text{size}_{i-1}/4 - 1$ items into the new, smaller table. Since $\text{num}_i = \text{size}_i/2 - 1$ after the operation has completed, $\alpha_i < 1/2$, and so

$$\begin{aligned}
\Phi_i &= \text{size}_i/2 - \text{num}_i \\
&= 1,
\end{aligned}$$

giving $\Delta\Phi_i = 1 - \text{size}_{i-1}/4$. Therefore, when the i th operation is a deletion that triggers a contraction, its amortized cost is

$$\begin{aligned}
\hat{c}_i &= c_i + \Delta\Phi_i \\
&= \text{size}_{i-1}/4 + (1 - \text{size}_{i-1}/4) \\
&= 1.
\end{aligned}$$

Finally, we handle the cases where the load factor fits one case of equation (16.5) before the operation and the other case afterward. We start with deletion, where we have $\text{num}_{i-1} = \text{size}_{i-1}/2$, so that $\alpha_{i-1} = 1/2$, beforehand, and $\text{num}_i = \text{size}_i/2 - 1$, so that $\alpha_i < 1/2$ afterward. Because $\alpha_{i-1} = 1/2$, we have $\Phi_{i-1} = 0$, and because $\alpha_i < 1/2$, we have $\Phi_i = \text{size}_i/2 - \text{num}_i = 1$. Thus we get that $\Delta\Phi_i = 1 - 0 = 1$. Since the i th operation is a deletion that does not cause a contraction, the actual cost c_i equals 1, and the amortized cost $\hat{c}_i$ is $c_i + \Delta\Phi_i = 1 + 1 = 2$.

Conversely, if the i th operation is an insertion that takes the load factor from below $1/2$ to equaling $1/2$, the change in potential $\Delta\Phi_i$ equals -1 . Again, the actual cost c_i is 1, and now the amortized cost $\hat{c}_i$ is $c_i + \Delta\Phi_i = 1 + (-1) = 0$.

In summary, since the amortized cost of each operation is bounded above by a constant, the actual time for any sequence of n operations on a dynamic table is $O(n)$.

Exercises

16.4-1

Using the potential method, analyze the amortized cost of the first table insertion.

16.4-2

You wish to implement a dynamic, open-address hash table. Why might you consider the table to be full when its load factor reaches some value α that is strictly less than 1? Describe briefly how to make insertion into a dynamic, open-address hash table run in such a way that the expected value of the amortized cost per

insertion is $O(1)$. Why is the expected value of the actual cost per insertion not necessarily $O(1)$ for all insertions?

16.4-3

Discuss how to use the accounting method to analyze both the insertion and deletion operations, assuming that the table doubles in size when its load factor exceeds 1 and the table halves in size when its load factor goes below $1/4$.

16.4-4

Suppose that instead of contracting a table by halving its size when its load factor drops below $1/4$, you contract the table by multiplying its size by $2/3$ when its load factor drops below $1/3$. Using the potential function

$$\Phi(T) = |2(T.num - T.size/2)|,$$

show that the amortized cost of a TABLE-DELETE that uses this strategy is bounded above by a constant.

Problems

16-1 Binary reflected Gray code

A **binary Gray code** represents a sequence of nonnegative integers in binary such that to go from one integer to the next, exactly one bit flips every time. The **binary reflected Gray code** represents a sequence of the integers 0 to $2^k - 1$ for some positive integer k according to the following recursive method:

- For $k = 1$, the binary reflected Gray code is $\langle 0, 1 \rangle$.
- For $k \geq 2$, first form the binary reflected Gray code for $k - 1$, giving the 2^{k-1} integers 0 to $2^{k-1} - 1$. Then form the reflection of this sequence, which is just the sequence in reverse. (That is, the j th integer in the sequence becomes the $(2^{k-1} - j - 1)$ st integer in the reflection). Next, add 2^{k-1} to each of the 2^{k-1} integers in the reflected sequence. Finally, concatenate the two sequences.

For example, for $k = 2$, first form the binary reflected Gray code $\langle 0, 1 \rangle$ for $k = 1$. Its reflection is the sequence $\langle 1, 0 \rangle$. Adding $2^{k-1} = 2$ to each integer in the reflection gives the sequence $\langle 3, 2 \rangle$. Concatenating the two sequences gives $\langle 0, 1, 3, 2 \rangle$ or, in binary, $\langle 00, 01, 11, 10 \rangle$, so that each integer differs from its predecessor by exactly one bit. For $k = 3$, the reflection of the binary reflected Gray code for $k = 2$ is $\langle 2, 3, 1, 0 \rangle$ and adding $2^{k-1} = 4$ gives $\langle 6, 7, 5, 4 \rangle$. Concatenating produces the sequence $\langle 0, 1, 3, 2, 6, 7, 5, 4 \rangle$, which in binary is $\langle 000, 001, 011, 010, 110, 111, 101, 100 \rangle$. In the binary reflected Gray code, only one bit flips even when wrapping around from the last integer to the first.

- a. Index the integers in a binary reflected Gray code from 0 to $2^k - 1$, and consider the i th integer in the binary reflected Gray code. To go from the $(i - 1)$ st integer to the i th integer in the binary reflected Gray code, exactly one bit flips. Show how to determine which bit flips, given the index i .
- b. Assuming that given a bit number j , you can flip bit j of an integer in constant time, show how to compute the entire binary reflected Gray code sequence of 2^k numbers in $\Theta(2^k)$ time.

16-2 Making binary search dynamic

Binary search of a sorted array takes logarithmic search time, but the time to insert a new element is linear in the size of the array. You can improve the time for insertion by keeping several sorted arrays.

Specifically, suppose that you wish to support SEARCH and INSERT on a set of n elements. Let $k = \lceil \lg(n + 1) \rceil$, and let the binary representation of n be $\langle n_{k-1}, n_{k-2}, \dots, n_0 \rangle$. Maintain k sorted arrays $A_0, A_1, \dots, A_{k-1}$, where for $i = 0, 1, \dots, k - 1$, the length of array A_i is 2^i . Each array is either full or empty, depending on whether $n_i = 1$ or $n_i = 0$, respectively. The total number of elements held in all k arrays is therefore $\sum_{i=0}^{k-1} n_i 2^i = n$. Although each individual array is sorted, elements in different arrays bear no particular relationship to each other.

- a. Describe how to perform the SEARCH operation for this data structure. Analyze its worst-case running time.
- b. Describe how to perform the INSERT operation. Analyze its worst-case and amortized running times, assuming that the only operations are INSERT and SEARCH.
- c. Describe how to implement DELETE. Analyze its worst-case and amortized running times, assuming that there can be DELETE, INSERT, and SEARCH operations.

16-3 Amortized weight-balanced trees

Consider an ordinary binary search tree augmented by adding to each node x the attribute $x.size$, which gives the number of keys stored in the subtree rooted at x . Let α be a constant in the range $1/2 \leq \alpha < 1$. We say that a given node x is **α -balanced** if $x.left.size \leq \alpha \cdot x.size$ and $x.right.size \leq \alpha \cdot x.size$. The tree as a whole is **α -balanced** if every node in the tree is α -balanced. The following amortized approach to maintaining weight-balanced trees was suggested by G. Varghese.

- a.** A $1/2$ -balanced tree is, in a sense, as balanced as it can be. Given a node x in an arbitrary binary search tree, show how to rebuild the subtree rooted at x so that it becomes $1/2$ -balanced. Your algorithm should run in $\Theta(x.size)$ time, and it can use $O(x.size)$ auxiliary storage.
- b.** Show that performing a search in an n -node α -balanced binary search tree takes $O(\lg n)$ worst-case time.

For the remainder of this problem, assume that the constant α is strictly greater than $1/2$. Suppose that you implement INSERT and DELETE as usual for an n -node binary search tree, except that after every such operation, if any node in the tree is no longer α -balanced, then you “rebuild” the subtree rooted at the highest such node in the tree so that it becomes $1/2$ -balanced.

We’ll analyze this rebuilding scheme using the potential method. For a node x in a binary search tree T , define

$$\Delta(x) = |x.left.size - x.right.size|.$$

Define the potential of T as

$$\Phi(T) = c \sum_{x \in T: \Delta(x) \geq 2} \Delta(x),$$

where c is a sufficiently large constant that depends on α .

- c.** Argue that any binary search tree has nonnegative potential and also that a $1/2$ -balanced tree has potential 0.
- d.** Suppose that m units of potential can pay for rebuilding an m -node subtree. How large must c be in terms of α in order for it to take $O(1)$ amortized time to rebuild a subtree that is not α -balanced?
- e.** Show that inserting a node into or deleting a node from an n -node α -balanced tree costs $O(\lg n)$ amortized time.

16-4 The cost of restructuring red-black trees

There are four basic operations on red-black trees that perform *structural modifications*: node insertions, node deletions, rotations, and color changes. We have seen that RB-INSERT and RB-DELETE use only $O(1)$ rotations, node insertions, and node deletions to maintain the red-black properties, but they may make many more color changes.

- a.** Describe a legal red-black tree with n nodes such that calling RB-INSERT to add the $(n + 1)$ st node causes $\Omega(\lg n)$ color changes. Then describe a legal

red-black tree with n nodes for which calling RB-DELETE on a particular node causes $\Omega(\lg n)$ color changes.

Although the worst-case number of color changes per operation can be logarithmic, you will prove that any sequence of m RB-INSERT and RB-DELETE operations on an initially empty red-black tree causes $O(m)$ structural modifications in the worst case.

- b.** Some of the cases handled by the main loop of the code of both RB-INSERT-FIXUP and RB-DELETE-FIXUP are *terminating*: once encountered, they cause the loop to terminate after a constant number of additional operations. For each of the cases of RB-INSERT-FIXUP and RB-DELETE-FIXUP, specify which are terminating and which are not. (*Hint*: Look at Figures 13.5, 13.6, and 13.7 in Sections 13.3 and 13.4.)

You will first analyze the structural modifications when only insertions are performed. Let T be a red-black tree, and define $\Phi(T)$ to be the number of red nodes in T . Assume that one unit of potential can pay for the structural modifications performed by any of the three cases of RB-INSERT-FIXUP.

- c.** Let T' be the result of applying Case 1 of RB-INSERT-FIXUP to T . Argue that $\Phi(T') = \Phi(T) - 1$.
- d.** We can break the operation of the RB-INSERT procedure into three parts. List the structural modifications and potential changes resulting from lines 1–16 of RB-INSERT, from nonterminating cases of RB-INSERT-FIXUP, and from terminating cases of RB-INSERT-FIXUP.
- e.** Using part (d), argue that the amortized number of structural modifications performed by any call of RB-INSERT is $O(1)$.

Next you will prove that there are $O(m)$ structural modifications when both insertions and deletions occur. Define, for each node x ,

$$w(x) = \begin{cases} 0 & \text{if } x \text{ is red ,} \\ 1 & \text{if } x \text{ is black and has no red children ,} \\ 0 & \text{if } x \text{ is black and has one red child ,} \\ 2 & \text{if } x \text{ is black and has two red children .} \end{cases}$$

Now redefine the potential of a red-black tree T as

$$\Phi(T) = \sum_{x \in T} w(x) ,$$

and let T' be the tree that results from applying any nonterminating case of RB-INSERT-FIXUP or RB-DELETE-FIXUP to T .

- f.* Show that $\Phi(T') \leq \Phi(T) - 1$ for all nonterminating cases of RB-INSERT-FIXUP. Argue that the amortized number of structural modifications performed by any call of RB-INSERT-FIXUP is $O(1)$.
- g.* Show that $\Phi(T') \leq \Phi(T) - 1$ for all nonterminating cases of RB-DELETE-FIXUP. Argue that the amortized number of structural modifications performed by any call of RB-DELETE-FIXUP is $O(1)$.
- h.* Complete the proof that in the worst case, any sequence of m RB-INSERT and RB-DELETE operations performs $O(m)$ structural modifications.

Chapter notes

Aho, Hopcroft, and Ullman [5] used aggregate analysis to determine the running time of operations on a disjoint-set forest. We'll analyze this data structure using the potential method in Chapter 19. Tarjan [430] surveys the accounting and potential methods of amortized analysis and presents several applications. He attributes the accounting method to several authors, including M. R. Brown, R. E. Tarjan, S. Huddleston, and K. Mehlhorn. He attributes the potential method to D. D. Sleator. The term “amortized” is due to D. D. Sleator and R. E. Tarjan.

Potential functions are also useful for proving lower bounds for certain types of problems. For each configuration of the problem, define a potential function that maps the configuration to a real number. Then determine the potential Φ_{init} of the initial configuration, the potential Φ_{final} of the final configuration, and the maximum change in potential $\Delta\Phi_{\text{max}}$ due to any step. The number of steps must therefore be at least $|\Phi_{\text{final}} - \Phi_{\text{init}}| / |\Delta\Phi_{\text{max}}|$. Examples of potential functions to prove lower bounds in I/O complexity appear in works by Cormen, Sundquist, and Wisniewski [105], Floyd [146], and Aggarwal and Vitter [3]. Krumme, Cybenko, and Venkataraman [271] applied potential functions to prove lower bounds on *gossiping*: communicating a unique item from each vertex in a graph to every other vertex.



Part V Advanced Data Structures

Introduction

This part returns to studying data structures that support operations on dynamic sets, but at a more advanced level than Part III. One of the chapters, for example, makes extensive use of the amortized analysis techniques from Chapter 16.

Chapter 17 shows how to augment red-black trees—adding additional information in each node—to support dynamic-set operations in addition to those covered in Chapters 12 and 13. The first example augments red-black trees to dynamically maintain order statistics for a set of keys. Another example augments them in a different way to maintain intervals of real numbers. Chapter 17 includes a theorem giving sufficient conditions for when a red-black tree can be augmented while maintaining the $O(\lg n)$ running times for insertion and deletion.

Chapter 18 presents B-trees, which are balanced search trees specifically designed to be stored on disks. Since disks operate much more slowly than random-access memory, B-tree performance depends not only on how much computing time the dynamic-set operations consume but also on how many disk accesses they perform. For each B-tree operation, the number of disk accesses increases with the height of the B-tree, but B-tree operations keep the height low.

Chapter 19 examines data structures for disjoint sets. Starting with a universe of n elements, each initially in its own singleton set, the operation UNION unites two sets. At all times, the n elements are partitioned into disjoint sets, even as calls to the UNION operation change the members of a set dynamically. The query FIND-SET identifies the unique set that contains a given element at the moment. Representing each set as a simple rooted tree yields surprisingly fast operations: a sequence of m operations runs in $O(m \alpha(n))$ time, where $\alpha(n)$ is an incredibly slowly growing function— $\alpha(n)$ is at most 4 in any conceivable application. The amortized analysis that proves this time bound is as complex as the data structure is simple.

The topics covered in this part are by no means the only examples of “advanced” data structures. Other advanced data structures include the following:

- **Fibonacci heaps** [156] implement mergeable heaps (see Problem 10-2 on page 268) with the operations INSERT, MINIMUM, and UNION taking only $O(1)$ actual and amortized time, and the operations EXTRACT-MIN and DELETE taking $O(\lg n)$ amortized time. The most significant advantage of these data structures, however, is that DECREASE-KEY takes only $O(1)$ amortized time. **Strict Fibonacci heaps** [73], developed later, made all of these time bounds actual. Because the DECREASE-KEY operation takes constant amortized time, (strict) Fibonacci heaps constitute key components of some of the asymptotically fastest algorithms to date for graph problems.
- **Dynamic trees** [415, 429] maintain a forest of disjoint rooted trees. Each edge in each tree has a real-valued cost. Dynamic trees support queries to find parents, roots, edge costs, and the minimum edge cost on a simple path from a node up to a root. Trees may be manipulated by cutting edges, updating all edge costs on a simple path from a node up to a root, linking a root into another tree, and making a node the root of the tree it appears in. One implementation of dynamic trees gives an $O(\lg n)$ amortized time bound for each operation, while a more complicated implementation yields $O(\lg n)$ worst-case time bounds. Dynamic trees are used in some of the asymptotically fastest network-flow algorithms.
- **Splay trees** [418, 429] are a form of binary search tree on which the standard search-tree operations run in $O(\lg n)$ amortized time. One application of splay trees simplifies dynamic trees.
- **Persistent** data structures allow queries, and sometimes updates as well, on past versions of a data structure. For example, linked data structures can be made persistent with only a small time and space cost [126]. Problem 13-1 gives a simple example of a persistent dynamic set.
- Several data structures allow a faster implementation of dictionary operations (INSERT, DELETE, and SEARCH) for a restricted universe of keys. By taking advantage of these restrictions, they are able to achieve better worst-case asymptotic running times than comparison-based data structures. If the keys are unique integers drawn from the set $\{0, 1, 2, \dots, u - 1\}$, where u is an exact power of 2, then a recursive data structure known as a **van Emde Boas tree** [440, 441] supports each of the operations SEARCH, INSERT, DELETE, MINIMUM, MAXIMUM, SUCCESSOR, and PREDECESSOR in $O(\lg \lg u)$ time. **Fusion trees** [157] were the first data structure to allow faster dictionary operations when the universe is restricted to integers, implementing these operations in $O(\lg n / \lg \lg n)$ time. Several subsequent data structures, including **exponential search trees** [17], have also given improved bounds on some or all of